

# Project 2 Report

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Course: CS791\_Robotics

Project: Train the Humanoid Robot to Walk and Run

## 1. Project Objective

Project 2 trains the Humanoid N1 robot to track commanded forward velocity across walking and running speeds while staying stable and robust. The policy is trained with PPO in Isaac Lab. Unlike Project 1, which focused on standing, this project adds locomotion commands and curriculum based training so the robot can produce coordinated walking and running behavior without falling.

## 2. Environment and Task Summary

The environment is implemented as a manager based RL environment in Isaac Lab. The robot is loaded from a USD model of Humanoid N1. The scene contains a flat plane terrain with contact sensing enabled for the robot bodies. The locomotion task uses the same 13 controlled joints from the standing project, which makes the action interface consistent with earlier work while extending the task to velocity tracking

### 2.1 Robot and Scene Configuration

Robot asset: Humanoid N1 loaded from N1\_main\_body\_raw.usd.

Terrain: flat plane.

Number of parallel environments: 4096 for training; 64 for play mode.

Environment spacing: 2.5 m in training, 3.0 m in play mode.

Contact sensor: enabled on all robot bodies, history length = 8, air-time tracking enabled.

Simulation time step:  $dt = 0.005$  s.

Control decimation: 4, so the policy acts every 0.02 s (50 Hz).

Episode length: 20 s

### 2.2 Policy observations (PolicyCfg)

The policy observation vector is the concatenation of:

- Base angular velocity (3D) in the base frame
- Projected gravity (3D) in the base frame
- Velocity commands (3D), generated from the base\_velocity command. In this project, the forward - velocity command is nonzero for walking and running, while lateral velocity and yaw rate are fixed to zero.
- Joint position relative offsets (13D) for the controlled joints
- Joint velocity relative values (13D) for the controlled joints

- Previous action (13D)

Therefore, the total policy observation dimension is:

$$3 + 3 + 3 + 13 + 13 + 13 = 48D$$

## 2.3 Critic observations (CriticCfgy)

The critic uses all policy observations plus additional privileged information:-

- Base linear velocity (3D)
- Base height (1D)
- Feet contact state (2D) using force thresholded contact information
- Feet speed in x-y (4D) for the left and right feet
- Feet height (2D)

So, the critic observation dimension is:

$$\text{Policy } 48D + \text{extras } (3 + 1 + 2 + 4 + 2) = 60D$$

The actor-critic design allows the critic to access richer state information during training while keeping the policy observation compact.

## 2.4 Action Space

The robot is controlled through 13 action joints, including both legs and the waist yaw joint:

- 6 joints for the left leg
- 6 joints for the right leg
- 1 waist yaw joint

Actions are implemented using `mdp.JointPositionActionCfgy`, where the action represents a normalized joint position command.

The configuration uses:

- `joint_names = ACTION_JOINTS_13`
- `scale = ACTION_SCALE_BY_JOINT`
- `use_default_offset = True`

The scale for each joint is computed from the safe symmetric range around the default pose using the predefined joint limits (`ACTIONS_MIN_13`, `ACTIONS_MAX_13`).

## 2.5 Command Space

The task uses a `base_velocity` command. In the current implementation, the command primarily varies the forward speed. Lateral velocity and yaw rate are fixed to zero. Commands are resampled every 10 seconds. A fraction of environments are assigned near zero standing commands, which helps preserve standing stability while the robot learns to walk and run.

| Command Component                       | Range          |
|-----------------------------------------|----------------|
| Forward velocity <code>lin_vel_x</code> | 0.0 to 3.2 m/s |

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| Lateral velocity lin_vel_y | 0.0 to 0.0 m/s                                    |
| Yaw rate ang_vel_z         | 0.0 to 0.0 rad/s                                  |
| Heading                    | -pi to pi (defined, but heading command disabled) |
| Resampling time            | 10.0 s                                            |
| Standing environments      | 20% initially, then reduced by curriculum         |

## 2.6 Reset and Termination Conditions

Reset and termination logic is necessary to keep training stable and to measure failure correctly.

| Category                       | Implementation                                                                                               |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|
| Reset: base state              | Randomized root pose and velocity; small x-y position offsets, small roll/pitch perturbations, random yaw.   |
| Reset: joints                  | Controlled joints reset close to default pose with scale range 0.95 to 1.05; joint velocities reset to zero. |
| Termination: time-out          | Episode ends after 20.0 s.                                                                                   |
| Termination: fall              | Triggered when base height drops below 0.42 m.                                                               |
| Termination: bad orientation   | Triggered when body tilt exceeds the configured angle limit of 1.0 rad.                                      |
| Termination: undesired contact | Triggered when the base link experiences strong contact force above threshold.                               |

## 3. Reward Design

The reward function combines tracking, posture, gait shaping, efficiency penalties, and failure penalties. The largest positive term is forward velocity tracking, while the largest negative term is termination. This makes successful survival and command following the main learning target.

| Reward Term         | Weight | Purpose                                                                               |
|---------------------|--------|---------------------------------------------------------------------------------------|
| track_lin_vel       | +6.0   | Main reward for tracking commanded forward velocity in the yaw frame.                 |
| track_ang_vel       | +3.0   | Yaw-rate tracking term; currently command is zero, so it stabilizes unwanted turning. |
| upright             | +1.0   | Keeps the torso aligned with gravity.                                                 |
| height              | +1.5   | Tracks nominal root height around 0.70 m.                                             |
| pose                | +0.25  | Keeps joints close to the default locomotion pose.                                    |
| stand_still         | +0.35  | Encourages stable near standing behavior for near-zero commands.                      |
| feet_air_time       | +0.6   | Encourages stepping/single support behavior when moving.                              |
| double_support_slow | +0.5   | Rewards stable support at low-speed commands.                                         |
| fly                 | -1.0   | Penalizes losing contact of both feet.                                                |
| base_z_lin_vel      | -0.25  | Reduces unnecessary vertical bouncing.                                                |

|                   |        |                                                            |
|-------------------|--------|------------------------------------------------------------|
| base_xy_ang_vel   | -0.08  | Penalizes excessive roll and pitch angular motion.         |
| joint_pos_limits  | -2.0   | Discourages motion near joint limits.                      |
| action_rate       | -0.01  | Smooths action changes.                                    |
| energy            | -5e-4  | Penalizes large joint torque usage.                        |
| feet_slide        | -0.25  | Penalizes foot slip during contact.                        |
| undesired_contact | -1.0   | Penalizes undesired contact, especially base collisions.   |
| waist_yaw_hold    | -0.4   | Discourages unnecessary waist yaw motion.                  |
| tracking_error    | -1.5   | Direct penalty on forward velocity error.                  |
| no_progress       | -2.0   | Penalizes failure to move when a nonzero command is given. |
| terminating       | -200.0 | Strong terminal penalty for failures.                      |

Both metrics `slip_metric` and `contact_ratio` come from the foot contact sensor. `slip_metric` uses the horizontal foot velocity during contact, while `contact_ratio` measures how many feet are in contact with the ground on average.

## 4. PPO Training Configuration

The training algorithm is PPO with an actor-critic architecture. The runner configuration uses a large number of environments and a short rollout horizon per environment, which is suitable for Isaac Lab locomotion training.

| Hyperparameter         | Value        | Comment                                              |
|------------------------|--------------|------------------------------------------------------|
| Algorithm              | PPO          | Actor-critic RL algorithm.                           |
| Random seed            | 42           | Fixed seed for reproducibility.                      |
| Device                 | cuda:0       | GPU training.                                        |
| Number of environments | 4096         | Massively parallel simulation.                       |
| Rollout length         | 64 steps/env | Equivalent to 1.28 s of policy time per environment. |
| Samples per iteration  | 262,144      | 4096 x 64 transitions before update.                 |
| Learning rate          | 3e-4         | Adaptive KL schedule used.                           |
| Learning epochs        | 8            | Optimization passes per rollout.                     |
| Mini-batches           | 16           | Mini-batch size is 16,384 samples.                   |
| Mini-batch size        | 16,384       | Computed from rollout buffer / 16.                   |
| Discount factor gamma  | 0.99         | Long-horizon return weighting.                       |
| GAE lambda             | 0.95         | Bias-variance tradeoff.                              |
| Entropy coefficient    | 0.001        | Maintains exploration.                               |
| Clip parameter         | 0.2          | Standard PPO objective clip.                         |
| Desired KL             | 0.01         | Adaptive learning rate schedule target.              |
| Max grad norm          | 1.0          | Gradient clipping.                                   |
| Value loss coefficient | 1.0          | Critic loss weight.                                  |

|                    |                 |                                                     |
|--------------------|-----------------|-----------------------------------------------------|
| Max iterations     | 6000            | Long training schedule for walk-to-run progression. |
| Save interval      | 50              | Checkpoint frequency.                               |
| Actor hidden dims  | [512, 256, 128] | ELU activation.                                     |
| Critic hidden dims | [512, 256, 128] | ELU activation.                                     |

Because the control interval is 0.02 s, a rollout of 64 control steps corresponds to 1.28 seconds of locomotion experience per environment. The episode length of 20 seconds corresponds to about 1000 control steps if the robot does not terminate early.

## 5. Curriculum Strategy

A curriculum is required for this project. In my implementation, the curriculum gradually changes the command velocity range, the standing environment ratio, and the push disturbance magnitude during training. The main design principle is to first learn standing and slow walking, then medium-speed walking, then high-speed velocity tracking, and finally specialize near the top running range. This staged design makes the locomotion task harder step by step while keeping training stable.

| Stage | Common Step Range | Forward Command Range | Standing Fraction | Push Disturbance                             |
|-------|-------------------|-----------------------|-------------------|----------------------------------------------|
| 1     | 0 – 32,000        | 0.0 - 0.8 m/s         | 0.25              | None                                         |
| 2     | 32,001 – 83,200   | 0.3 - 1.5 m/s         | 0.10              | Mild pushes: $x \pm 0.10$ , $y \pm 0.08$     |
| 3     | 83,201 – 153,600  | 1.0 - 2.4 m/s         | 0.05              | Moderate pushes: $x \pm 0.15$ , $y \pm 0.10$ |
| 4     | 153,601 – 256,000 | 1.8 - 3.2 m/s         | 0.02              | No pushes; prioritize speed tracking         |
| 5     | 256,001+          | 2.5 - 3.2 m/s         | 0.00              | No pushes; specialize for running            |

It reflects a practical tradeoff: pushes are used mainly during walking stages for robustness, while the highest-speed stages remove pushes so the policy can focus on learning a fast gait without being dominated by disturbances.

## 6. Robustness

The project requires at least two robustness conditions. The current implementation already includes two strong candidates:

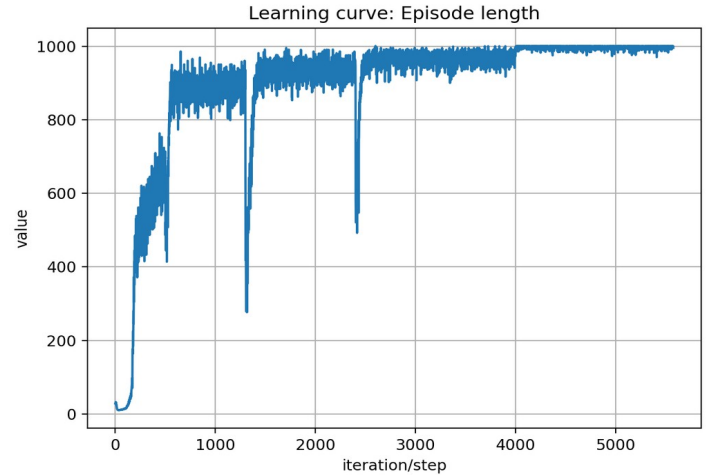
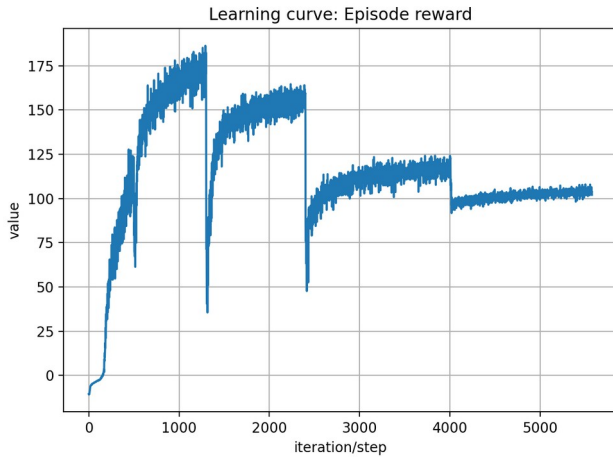
- Disturbance pushes are applied every 10-15 seconds. During the walking curriculum stages, the push velocity range is gradually increased up to  $x = \pm 0.15$  m/s and  $y = \pm 0.10$  m/s. Pushes are disabled in the high-speed running stages so the policy can focus on speed tracking.
- Domain randomization is applied at startup. Robot body material properties are randomized with static friction in [0.6, 1.0], dynamic friction in [0.4, 0.8], and restitution in [0.0, 0.005]. The

base-link mass is randomized by adding a value in  $[-1.0, 1.0]$  kg.

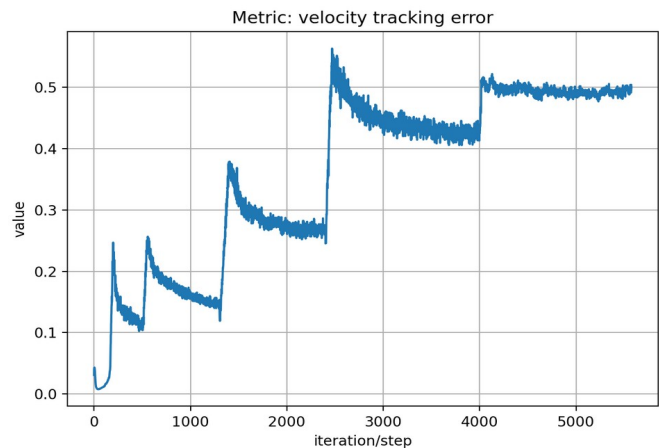
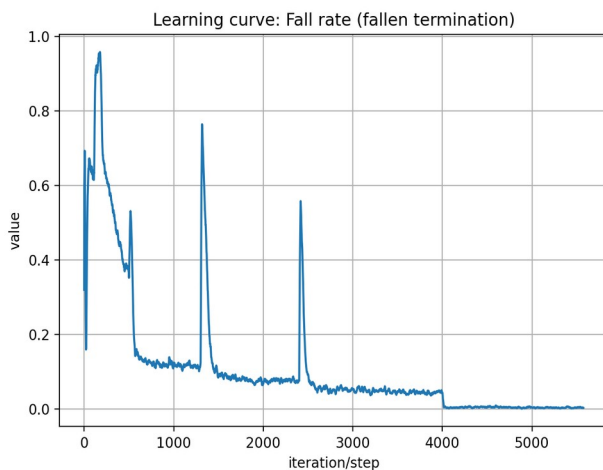
Robustness is tested with both domain randomization, disturbance pushes and reset enabled. During evaluation, friction/base mass randomization remains active, and random pushes are applied every 10-15 seconds. I report success rate, tracking error, slip/contact metrics, and termination breakdown under this combined robustness setting.

## 7. Logging and Required Plots

The project requires plots for episode reward, episode length, fall rate, tracking error, reward breakdown, and at least one gait/contact metric. Below are the required plots.

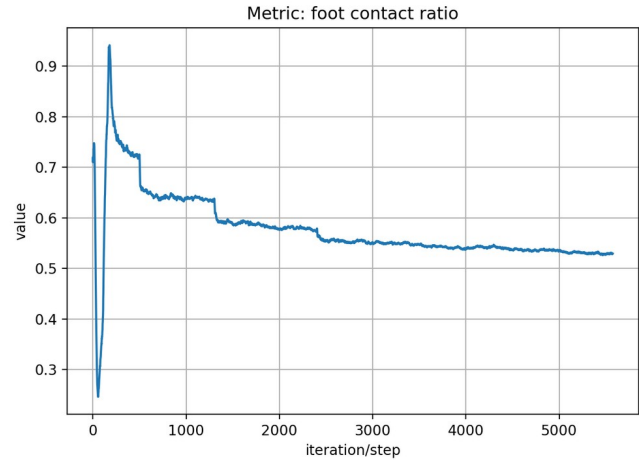
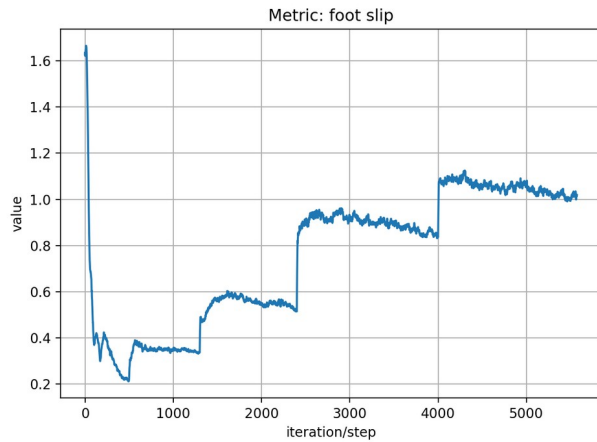


The learning curves show that the policy gradually learned stable locomotion. Episode length increased to nearly 1000 steps, which means the robot fell less often and could walk or run for whole 20 second episode. The large drops around about 1300, 2400, and 4000 iterations are reasonable because the curriculum made the task harder at those stages. After each drop, the curves recovered again, which shows that the policy adapted to the new speed ranges and training conditions.

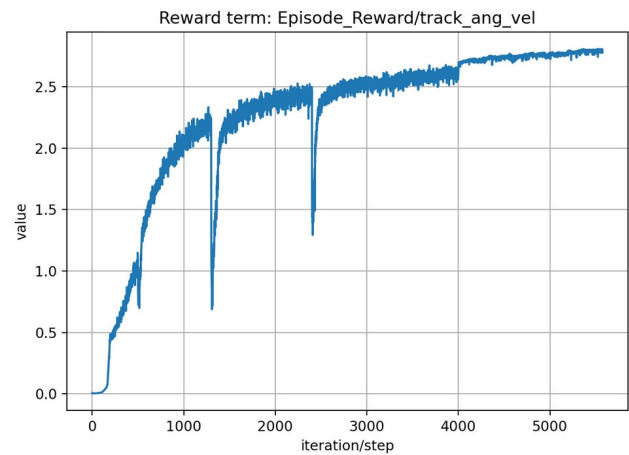
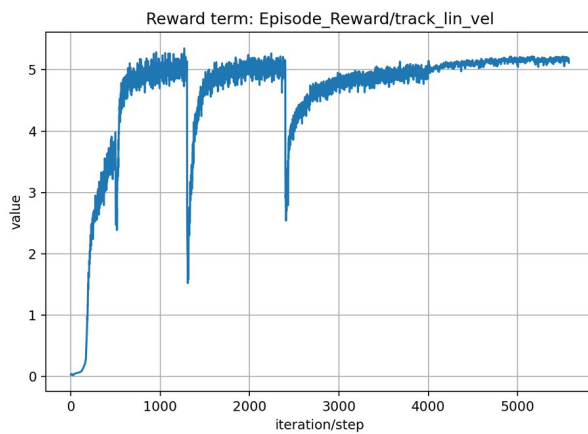


The fall rate decreased over training and became close to zero near the end, which shows better stability and fewer falls. The spikes are expected because the curriculum made the task harder at later stages. The velocity tracking error increased when higher speeds were introduced, but it later became more

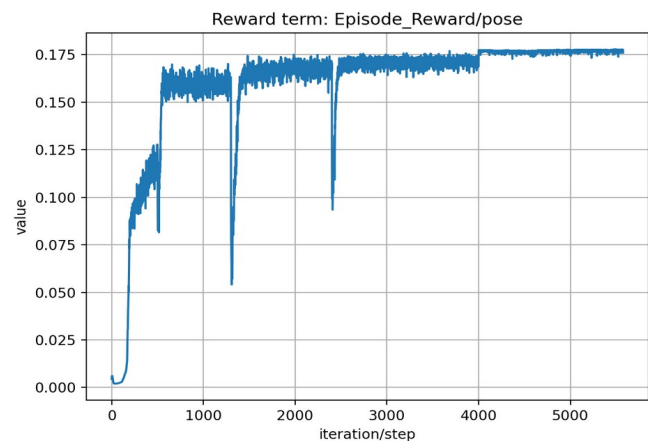
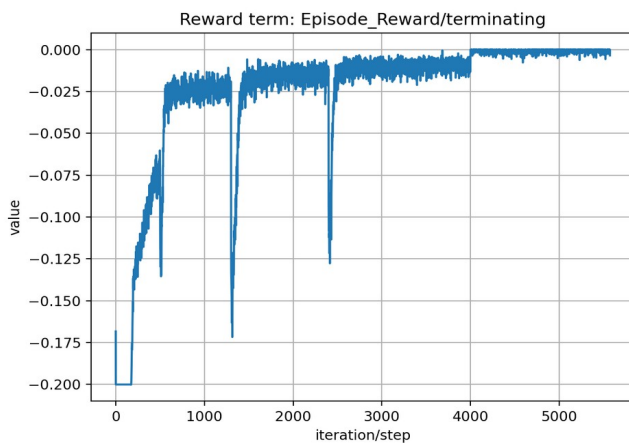
stable. This suggests that the policy adapted to harder walking and running commands.



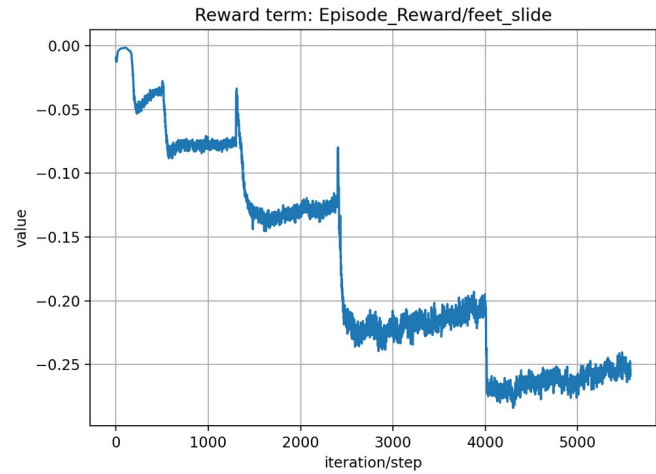
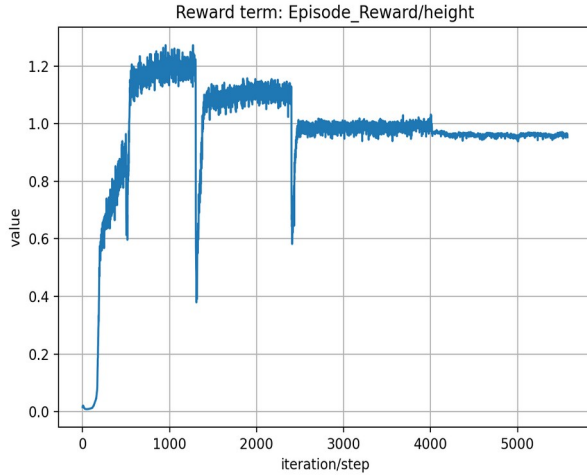
The foot slip metric becomes higher in later stages, which suggests that faster walking and running made the task harder and caused more sliding. The foot contact ratio slowly decreases over training, which is reasonable because the robot spends less time with both feet on the ground as it moves from slower walking to faster locomotion.



The track\_lin\_vel reward increased over training, which means the robot became better at following the commanded linear forward velocity. The track\_ang\_vel reward also increased, which suggests that the robot reduced unwanted yaw rotation and maintained a more stable heading, because the commanded yaw rate was fixed to zero. The drops in both curves are expected because the curriculum made the task harder at later stages

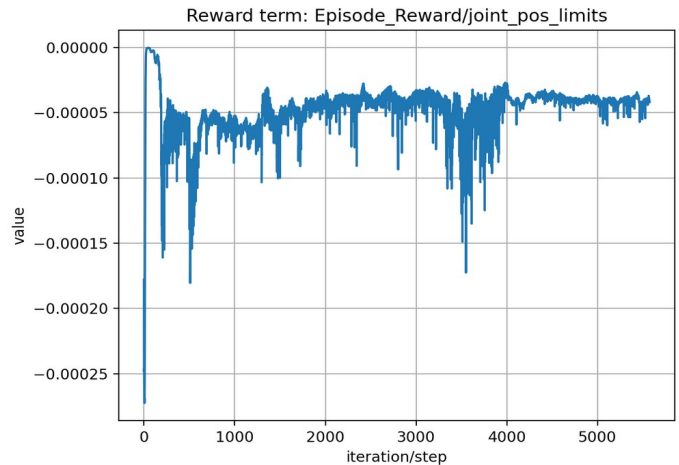
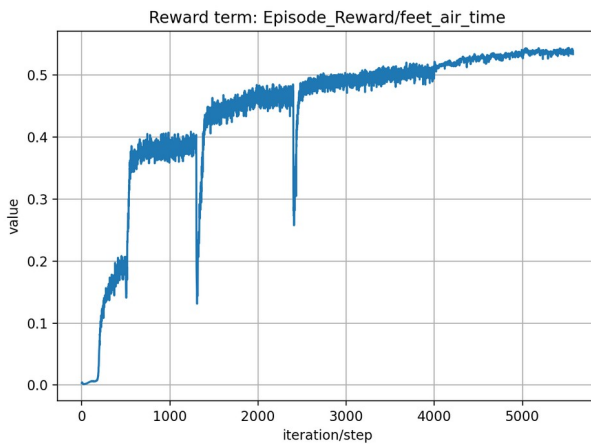


The terminating reward becomes less negative over training, which means the robot falls less often and receives fewer termination penalties. The pose reward also increases, which shows that the robot keeps a more stable and consistent body posture during locomotion.



The height reward is higher in the early stages, when the task focuses more on slow walking and the robot can keep its base height closer to the target. In later stages, when the curriculum moves to faster walking and running, the height reward becomes slightly lower but stays stable, which suggests that the robot still keeps a reasonable body height under harder commands.

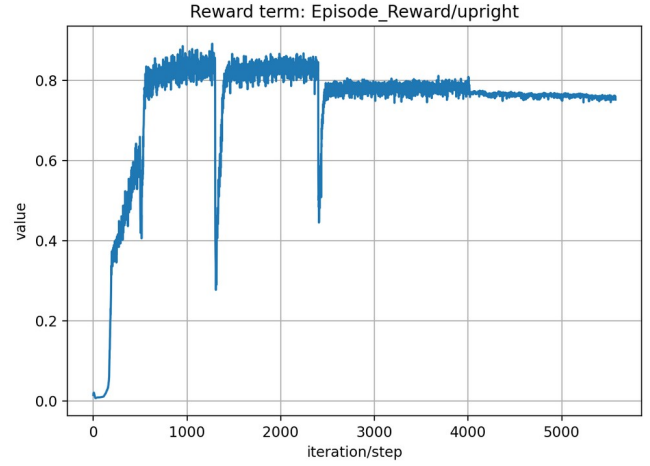
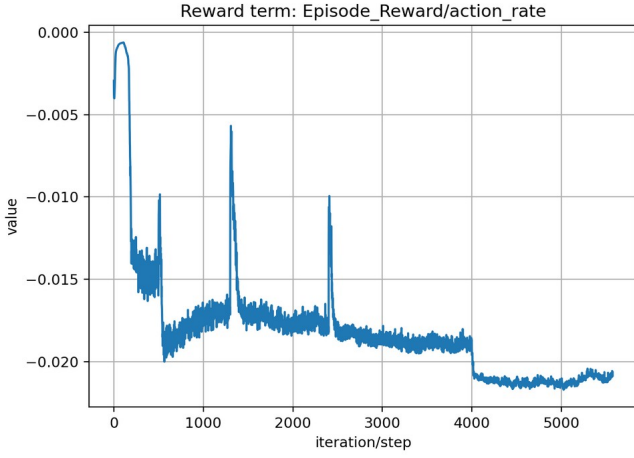
The feet\_slide reward becomes more negative from one stage to the next, especially after the curriculum moves to faster locomotion stages. This suggests that foot sliding increases at higher speeds, which is reasonable because the robot has to move more aggressively in the later stages.



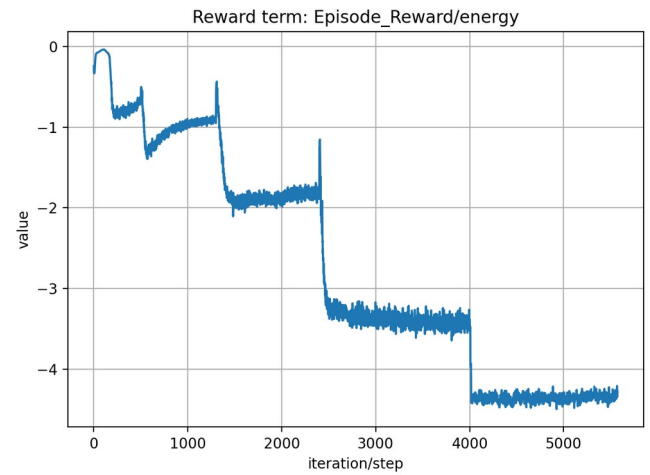
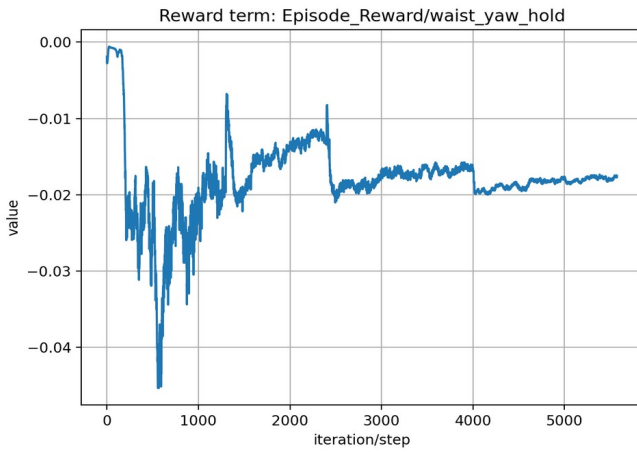
The feet\_air\_time reward increases over training, which shows that the robot learned more active stepping behavior. The drops are likely caused by curriculum changes, when the command range becomes harder.

The joint\_pos\_limits penalty stays very close to zero during most of training. This suggests that the robot usually keeps its joints within safe limits, although there are some temporary spikes when the task becomes harder.





The spikes are likely caused by curriculum stage changes, when the command range becomes harder and the policy needs time to adapt. The action\_rate reward becomes more negative in later stages, which suggests that the robot uses larger or faster action changes when the curriculum moves to faster locomotion. The upright reward stays high during training, which shows that the robot keeps a mostly stable upper-body posture even as the task becomes harder.



The waist\_yaw\_hold reward stays only slightly negative in later stages, which suggests that the robot keeps the waist yaw more controlled after early unstable training. The energy reward becomes more negative across stages, which means the robot uses more effort as the curriculum moves from slow walking to faster locomotion.

## 8. Evaluation Protocol

The policy was evaluated at three representative commanded speeds: 1.5 m/s, 2.5 m/s, and 3.2 m/s, covering walking, fast walking, and running. The result was evaluated by running python script file in the IsaacLab environment. The results were saved as csv file. These speeds were selected to satisfy the requirement of evaluating at least two walking speeds and one running speed.

All commanded-speed tracking results below were evaluated under a combined robustness setting with two enabled conditions: push disturbance and domain randomization.

| Evaluation Condition | Forward Command |
|----------------------|-----------------|
| Walk                 | 1.5 m/s         |
| Fast walk            | 2.5 m/s         |
| Run                  | 3.2 m/s         |

### 8.1 Main Evaluation Results Table

| Commanded Speed (m/s) | Success Rate (%) | Mean $v_x$ (m/s) | Mean error $ v_x - v_{x\_cmd} $ (m/s) | Mean Episode Length (s) | Slip Metric |
|-----------------------|------------------|------------------|---------------------------------------|-------------------------|-------------|
| 1.5                   | 100              | 1.545            | 0.0112                                | 20                      | 0.4394      |
| 2.5                   | 100              | 2.404            | 0.118                                 | 20                      | 0.8883      |
| 3.2                   | 86               | 2.98             | 0.262                                 | 19                      | 1.4835      |

### 8.2 Termination Breakdown

| Commanded Speed (m/s) | Time-out | Fall | Bad Orientation | Undesired Contact |
|-----------------------|----------|------|-----------------|-------------------|
| 1.5                   | 1.0      | 0.0  | 0.0             | 0.0               |
| 2.5                   | 1.0      | 0.0  | 0.0             | 0.0               |
| 3.2                   | 0.86     | 0.11 | 0.03            | 0.0               |

### 8.3 Robustness Results

| Robustness Condition                    | Commanded Speed(s) | Success Rate (%) | Tracking Error $ v_x - v_{x\_cmd} $ (m/s) | Key Observation                                                                                |
|-----------------------------------------|--------------------|------------------|-------------------------------------------|------------------------------------------------------------------------------------------------|
| Push disturbance + Domain randomization | 1.5                | 100              | 0.0112                                    | Achieve velocity, Nearly natural human-like gait                                               |
| Push disturbance + Domain randomization | 2.5                | 100              | 0.118                                     | Achieve velocity, Not natural human-like gait                                                  |
| Push disturbance + Domain randomization | 3.2                | 86               | 0.262                                     | Some failure cases reduce mean velocity, but high successful rate, Not natural human-like gait |

## 9. Analysis and Discussion

The implemented environment is well aligned with the Project 2 requirements. It includes a command-tracking task, reset and termination logic, a staged curriculum, disturbance pushes, domain randomization, and logging terms for slip and foot contact behavior. The PPO configuration is also suitable for large locomotion learning.

Because yaw rate and lateral commands are fixed to zero in the present implementation, this report focuses only on forward locomotion. This is acceptable for the current project scope.

In normal walking at 1.5 m/s, the robot performs very well. It tracks the commanded speed closely, stays upright, and completes the full evaluation window without falling. This shows that the policy learned stable and reliable walking behavior.

The results also show good command tracking at medium speeds and reasonable performance at fast walking speeds. However, performance drops at 3.2 m/s, which is the most difficult running condition in this project. Under the combined robustness setting with push disturbance and domain randomization,

the 3.2 m/s running command achieved an 86% success rate with a tracking error of 0.262. This shows that the policy can complete most high-speed episodes, but gait quality remains poor and some failure cases still occur.

## 10. Failure Modes and Possible Improvements

- High-speed running may succeed in speed tracking but still show poor gait aesthetics, such as nonparallel feet or torso sway.
- The final curriculum stages specialize strongly toward 2.5 to 3.2 m/s, which may reduce performance at slow walking if not monitored.
- Because terrain is a plane, the policy may still be sensitive to slopes or uneven terrain unless additional tests are added.
- The observation does not include explicit phase information, so gait regularity must emerge indirectly from rewards and contact terms.

I spent a lot of time tuning the parameters for a more human-like gait, but the result was still limited. So I suggest some potential improvements for future work:

- Future work can include training on more challenging terrain, such as small slopes and slightly uneven ground.
- Another possible improvement is to manually tune the reward weights to better balance speed tracking and gait stability.
- Another direction is to use human motion references or imitation learning so the robot can produce a more natural walking and running gait.
- Methods such as DeepMimic-style training, Adversarial Motion Priors (AMP), or diffusion-based motion policies could also be explored to improve gait naturalness.

## 11. Conclusion

This report presents a successful Project 2 locomotion setup for humanoid N1 that extends the previous standing task into walk and run command tracking. The implementation uses a 13-DOF action interface, asymmetric observations, a reward function, PPO training, and a five stage curriculum. It also includes two robustness mechanisms required by the assignment: pushes and domain randomization. The results show that the policy can achieve stable forward locomotion across walking and running speeds, although performance slightly drops at the highest speed and the gait is still not fully natural. Overall, the project demonstrates that the robot can track forward commands with reasonable robustness, while still leaving room for future improvement in gait quality.